Amit Nishad

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.94
Sigma School of Excellence, Jhalrapatan, Jhalawar, Rajasthan	Class XII (CBSE)	2024	88.60%
Jawahar Navodaya Vidyalaya (JNV), Saraipali, Mahasamund, Chhattisgarh	Class X (CBSE)	2022	91.80%

PROJECTS

MentoraAI | AI-Powered RAG Document Assistant | GitHub | Live Demo
Tech Stack: Python, FastAPI, React, Vite, LangChain, Google Gemini, ChromaDB, Docker, Render, Vercel

- Built an end-to-end **Retrieval-Augmented Generation (RAG)** application enabling users to upload PDF documents and ask natural language questions with AI-generated, source-cited responses.
- Engineered a modular **FastAPI** backend implementing document ingestion, text chunking, embedding generation, semantic retrieval, and RESTful APIs for document management and conversational querying.
- Integrated **LangChain, Gemini Embeddings, Gemini Flash, and ChromaDB** to perform **vector-based semantic search**, retrieve relevant document chunks, and generate context-grounded answers with page-level citations.
- Developed a responsive **React + Vite** frontend featuring PDF uploads, AI chat, document management, Markdown rendering, and seamless backend integration using Axios.
- Containerized the backend using **Docker** and deployed the application on **Render** with the frontend hosted on **Vercel**, configuring production-ready environment variables, CORS, and debugging deployment issues.

Medisense | AI-Powered Medical Record Management System | GitHub
Tech Stack: React.js, FastAPI, PostgreSQL, OpenAI API, Scikit-learn, Docker, AWS (RDS, S3)

- Built a **full-stack medical record management platform** enabling patient management, medical history tracking, and secure document handling using **React.js, FastAPI, and PostgreSQL**.
- Trained a **Logistic Regression** model using **Scikit-learn and StandardScaler** for diabetes-risk prediction, achieving **82.6% ROC-AUC** on the Pima Indians Diabetes Dataset, with prediction history persisted to **PostgreSQL**.
- Integrated the **OpenAI API** to auto-generate **AI-based medical summaries** by combining structured patient records with unstructured text extracted from PDF documents using **PyMuPDF**.
- Designed a **normalized 4-table PostgreSQL schema** (patients, records, documents, predictions) and built **RESTful APIs** consumed by a **Tailwind CSS-styled React frontend**.
- Containerized the application using **Docker & Docker Compose** and deployed backend infrastructure to **AWS (RDS, S3)**, with **ECS, ALB, and CI/CD via GitHub Actions** in progress.

ConvoX | Real-Time Video Conferencing Platform | GitHub | Live Demo
Tech Stack: React, Vite, Tailwind CSS, Node.js, Express.js, MongoDB, Socket.IO, WebRTC, JWT, Vercel, Render

- Built and deployed a full-stack **real-time video conferencing platform** supporting multi-user rooms, peer-to-peer audio/video communication, real-time chat, and microphone/camera controls using **WebRTC and Socket.IO**.
- Engineered the **WebRTC signaling architecture** using Socket.IO for room management, SDP offer/answer exchange, and ICE candidate signaling, enabling direct **peer-to-peer media streaming** between participants.
- Developed **JWT-based authentication** with bcrypt and MongoDB Atlas, implementing RESTful APIs for user registration/login and integrating the React frontend with the backend using Axios.
- Deployed the **React/Vite frontend on Vercel** and **Node.js/Express backend on Render**, configuring production environment variables, CORS, SPA routing, and resolving deployment-specific issues.

SKILLS

Languages: Java, Python, C, JavaScript

Frontend: React.js, HTML5, CSS3, Tailwind CSS, Vite

Backend: Node.js, Express.js, REST APIs, JWT

Databases: MongoDB, PostgreSQL, SQL

AI/ML & GenAI: Machine Learning, Scikit-learn, LangChain, Retrieval-Augmented Generation (RAG), OpenAI API, Google Gemini API, Vector Embeddings, ChromaDB

DevOps & Cloud: Docker, Docker Compose, Git, GitHub, AWS (RDS, S3), Render, Vercel

Core CS Concepts: Data Structures & Algorithms, OOP, Operating Systems, Computer Networks, MVC Architecture,